

Literature Status: Sierpiński Sets Are Not Productively Hurewicz under CH

Run fa_banach_001 — lane 1

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Status: literature already answered. Problem 5.5 of arXiv:1607.01687 is answered affirmatively, in a strictly stronger form, by Theorem 3.1 of arXiv:2503.13146. The later authors explicitly identify their theorem as an answer to this exact problem.

1 Original question

P. Szewczak and B. Tsaban, *Products of general Menger spaces*, arXiv:1607.01687, later published in *Topology and its Applications* **255** (2019), 41–55, pose the following on arXiv PDF page 13 [1].

Source Problem 5.5. *Does the Continuum Hypothesis imply that no Sierpiński set is productively Hurewicz?*

Here a space S is productively Hurewicz if $S \times H$ is Hurewicz for every Hurewicz space H .

2 Separate later answer

W. Marciszewski, R. Pol, and P. Zakrzewski, *On Sierpiński sets, Hurewicz spaces and Hilgers functions*, arXiv:2503.13146, published in *Annals of Pure and Applied Logic* **177** (2026), 103680, state immediately before Theorem 3.1 on arXiv PDF page 6 that their result gives “a positive answer (in a strong form)” to Problem 7.5 of their reference [ST1], repeated as Problem 5.5 of [ST2]. Their [ST2] is precisely the source paper above. The theorem is as follows [2].

Later theorem. *For every Sierpiński set $S \subseteq 2^{\aleph_1}$ of cardinality at least $\mathfrak{b}$, there is a Hurewicz λ' -set $H \subseteq 2^{\aleph_1}$ such that $S \times H$ is not Hurewicz. If $\mathfrak{b} = \mathfrak{d}$, the factor H may be chosen so that $S \times H$ is not even Menger.*

Under CH, every Sierpiński set has cardinality $\mathfrak{c} = \aleph_1 = \mathfrak{b} = \mathfrak{d}$. Theorem 3.1 therefore applies to every Sierpiński set S and supplies a Hurewicz factor H for which $S \times H$ is not Hurewicz (indeed, not Menger). Thus no Sierpiński set is productively Hurewicz, exactly as Problem 5.5 asks.

3 Scope and search status

This packet records a complete later-literature answer to Problem 5.5 only. The source also poses Problems 5.2–5.4. A bounded exact-label and core-phrase search through 13 August 2026 did not locate comparably explicit later resolutions of those three problems. Direct checks of their most

immediate routes encounter, respectively, the need to preserve Lindelöfness in products with general spaces and the gap between a forcing consistency construction of a Menger cofinite-filter scale and a consequence of CH alone. No claim about the current status of Problems 5.2–5.4 is made here.

No new mathematical result is claimed in this packet.

References

- [1] P. Szewczak and B. Tsaban, *Products of general Menger spaces*, Topology Appl. **255** (2019), 41–55; arXiv:1607.01687; DOI: 10.1016/j.topol.2019.01.005.
- [2] W. Marciszewski, R. Pol, and P. Zakrzewski, *On Sierpiński sets, Hurewicz spaces and Hilgers functions*, Ann. Pure Appl. Logic **177** (2026), 103680; arXiv:2503.13146; DOI: 10.1016/j.apal.2025.103680.